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Find a basis and calculate the dimension for the following subspace of 2×2 matrices (with standard addition and multiplication by a scalar operations)

$$\text{span} \left\{ \begin{pmatrix} 3 & 4 \\ 6 & 7 \end{pmatrix}, \begin{pmatrix} 2 & 3 \\ 2 & 2 \end{pmatrix}, \begin{pmatrix} -3 & -6 \\ 0 & -6 \end{pmatrix}, \begin{pmatrix} 3 & 9 \\ -7 & 10 \end{pmatrix} \right\}$$

Enter up to 4 basis matrices. For any matrices not required in the basis, enter $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

$$\begin{pmatrix} \boxed{3} & \boxed{4} \\ \boxed{6} & \boxed{7} \end{pmatrix}, \begin{pmatrix} \boxed{2} & \boxed{3} \\ \boxed{2} & \boxed{2} \end{pmatrix}, \begin{pmatrix} \boxed{-3} & \boxed{-6} \\ \boxed{0} & \boxed{-6} \end{pmatrix}, \begin{pmatrix} \boxed{3} & \boxed{9} \\ \boxed{-7} & \boxed{10} \end{pmatrix}$$

What is the dimension of the subspace?

Correct. Your basis matrices are independent and span the entire subspace. You were awarded  5 marks.

You scored 5 marks for this part.

Score: 5/5 

Answered

The vector space spanned by the following set of polynomials is a subspace of polynomials of degree $n \leq 5$.

$$\{3x^3 - 2x^2 - 2x^4 + 2x^5, \quad 2x^2 - 3x^3 + 4x^4 + 2x^5, \quad 3x^2 - 6x^3 - 2x^4 - 10x^5, \quad 3x^2 + 6x^4 - 3x^3\}$$

Identify a basis of the subspace. Enter the coefficients in a matrix such that

Rows: Columns:

$$\begin{pmatrix} \begin{matrix} -2 & 3 & -2 & 2 \end{matrix} \\ \begin{matrix} 2 & -3 & 4 & 2 \end{matrix} \\ \begin{matrix} 3 & -6 & -2 & -10 \end{matrix} \end{pmatrix} \begin{pmatrix} x^2 \\ x^3 \\ x^4 \\ x^5 \end{pmatrix} = \begin{pmatrix} \text{basis polynomials} \end{pmatrix}$$

(reduce the size of the matrix if the dimension is less than 4)

What is the dimension of the subspace?

Correct. Your basis polynomials are independent and span the entire subspace. You were awarded **2** marks. 

You scored **2** marks for this part.

Score: 2/2 

Answered

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Consider the set of ordered pairs defined on the Cartesian product $V = \{0, 1, 2\} \times \{0, 1, 2\}$, i.e. pairs of the form $\mathbf{x} = (x_1, x_2) \in V$ with $x_1 \in \{0, 1, 2\}$ and $x_2 \in \{0, 1, 2\}$. We can also refer to the pairs as 2-dimensional vectors.

We define scalar multiplication of elements in V such that for a scalar $a \in \{0, 1, 2\}$, we have $a \otimes \mathbf{x} = (a \otimes x_1, a \otimes x_2)$. Furthermore, vector addition between elements in V is defined such that $\mathbf{x} \oplus \mathbf{y} = (x_1 \oplus y_1, x_2 \oplus y_2)$. The addition ($\oplus$) and multiplication ($\otimes$) operations for scalars are defined according to the following tables.

$\oplus$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\otimes$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

The set of vectors in V along with $\otimes$ and $\oplus$ defines a vector space. The following will ask you to provide examples for each property (see p328 of the text). Note that the examples you'll provide don't **prove** that the properties are satisfied, but rather allow you to get an intuitive feel for how the properties work. The standard symbol '+' will be used to mean $\oplus$ in each of the axioms.

a)

First, give three distinct examples for vectors in V . Do not include $(0, 0)$.

$$\mathbf{u} = \begin{pmatrix} \boxed{1} & \boxed{0} \end{pmatrix}$$

$$\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix}$$

$$\mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix}$$

Submit part

Proceed to the next part. You were awarded 3 marks. ✓

You scored 3 marks for this part.

Score: 3/3 ✓

Answered

b)

Axioms for Vector Addition

For each of the following, fill the gaps for the examples. Note that $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ should correspond with the vectors you provided in part a (write these down on paper to save yourself scrolling up and down).

A1. If $\mathbf{u}$ and $\mathbf{v}$ are in V , then $\mathbf{u} + \mathbf{v}$ is in V .

Example: $\mathbf{u}$ and $\mathbf{v}$ are in V , the vector $\mathbf{u} + \mathbf{v} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix}$  is also in V .

A2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ for all $\mathbf{u}, \mathbf{v}$ in V .

Example: $\mathbf{u} + \mathbf{v} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix}$ , $\mathbf{v} + \mathbf{u} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix}$ 

A3. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ for all $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ in V .

Example: $\mathbf{v} + \mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{2} \end{pmatrix}$ , $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = \begin{pmatrix} \boxed{2} & \boxed{2} \end{pmatrix}$ 
 $\mathbf{u} + \mathbf{v} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix}$ , $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \begin{pmatrix} \boxed{2} & \boxed{2} \end{pmatrix}$ 

A4. An element $\mathbf{0}$ in V exists such that $\mathbf{v} + \mathbf{0} = \mathbf{v} = \mathbf{0} + \mathbf{v}$ for every $\mathbf{v}$ in V .

Example: The zero element is $\mathbf{0} = \begin{pmatrix} \boxed{0} & \boxed{0} \end{pmatrix}$ .

A5. For each $\mathbf{v}$ in V , an element $-\mathbf{v}$ in V exists such that $-\mathbf{v} + \mathbf{v} = \mathbf{0}$ and $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.

Example: $-\mathbf{u} = \begin{pmatrix} \boxed{2} & \boxed{0} \end{pmatrix}$ , and $-\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix}$ .

Gap 0

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded 1 mark. ✓

Gap 1

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded 1 mark. ✓

Gap 2

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded 1 mark. ✓

Gap 3

Your answer is correct. You were awarded 1 mark. ✓

Gap 4

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded 1 mark. ✓

Gap 5

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded 1 mark. ✓

Gap 6

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded 1 mark. ✓

Gap 7

Your answer is correct. You were awarded 1 mark. ✓

Gap 8

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded 1 mark. ✓

Gap 9

Your answer is correct. You were awarded 1 mark. ✓

You scored 10 marks for this part.

Score: 10/10 ✓

Answered

c)

Now enter 2 values, a, b , with both taken from $\{0, 1, 2\}$. They do not have to be different but ensure that $a \neq 0$.

$a =$

$b =$

Submit part

Proceed to the next part. You were awarded **2** marks. 

You scored **2** marks for this part.

Score: 2/2 

Answered

d)

Axioms for Scalar Multiplication

As with the addition axioms, fill out the examples below using your values of a and b , as well as the same $\mathbf{u}, \mathbf{v}, \mathbf{w}$ from part a. (write them all down to save you scrolling up)

Note that $a + b$ is defined the same as for $a \oplus b$ given in the table, and that in contrast to the axioms of page 328 of the text, here we assume a, b are in $\{0, 1, 2\}$ rather than in $\mathbb{R}$ (this structure is called a 'Field').

S1. If $\mathbf{v}$ is in V , then $a\mathbf{v}$ is in V for all a in $\{0, 1, 2\}$.

Example: $a\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix}$ ✓, which is in V .

S2 $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$ for all $\mathbf{v}$ and $\mathbf{w}$ in V and all a in $\{0, 1, 2\}$.

Example: $\mathbf{v} + \mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{2} \end{pmatrix}$ ✓, $a(\mathbf{v} + \mathbf{w}) = \begin{pmatrix} \boxed{1} & \boxed{2} \end{pmatrix}$ ✓,
 $a\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix}$ ✓, $a\mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix}$ ✓,
 $a\mathbf{v} + a\mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{2} \end{pmatrix}$ ✓.

S3 $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$ for all $\mathbf{v}$ in V and all a, b in $\{0, 1, 2\}$.

Example: $(a + b) = \boxed{0}$ ✓, $(a + b)\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{0} \end{pmatrix}$ ✓,
 $a\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix}$ ✓, $b\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix}$ ✓,
 $a\mathbf{v} + b\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{0} \end{pmatrix}$ ✓.

S4 $a(b\mathbf{v}) = (ab)\mathbf{v}$ for all $\mathbf{v}$ in V and all a, b in $\{0, 1, 2\}$.

Example: $b\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix}$ ✓, $a(b\mathbf{v}) = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix}$ ✓,
 $ab = \boxed{2}$ ✓, $(ab)\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix}$ ✓.

S5 $1\mathbf{v} = \mathbf{v}$ for all $\mathbf{v}$ in V .

Example: $1\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix}$ ✓.

Gap 0

Your answer is correct. You were awarded 1 mark. ✓

Gap 1

Your answer is correct. You were awarded 1 mark. ✓

Gap 2

Your answer is correct. You were awarded 1 mark. ✓

Gap 3

Your answer is correct. You were awarded 1 mark. ✓

Gap 4

Your answer is correct. You were awarded 1 mark. ✓

Gap 5

Your answer is correct. You were awarded 1 mark. ✓

Gap 6

Your answer is correct. You were awarded 1 mark. ✓

Gap 7

Your answer is correct. You were awarded 1 mark. ✓

Gap 8

Your answer is correct. You were awarded 1 mark. ✓

Gap 9

Your answer is correct. You were awarded 1 mark. ✓

Gap 10

Your answer is correct. You were awarded 1 mark. ✓

Gap 11

Your answer is correct. You were awarded 1 mark. ✓

Gap 12

Your answer is correct. You were awarded 1 mark. ✓

Gap 13

Your answer is correct. You were awarded 1 mark. ✓

Gap 14

Your answer is correct. You were awarded 1 mark. ✓

Gap 15

Your answer is correct. You were awarded 1 mark. ✓

You scored 16 marks for this part.

Score: 16/16 ✓

Answered

Submit all parts

Score: 31/31 ✓

Try another question like this one

Reveal answers

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The following two matrices are related by a sequence of elementary row operations:

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 & 1 \\ 1 & 2 & -2 & 3 & 0 \\ 2 & 4 & 1 & 1 & -2 \\ 1 & 2 & -2 & 3 & -3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Let W be a subspace of $\mathbf{P}_3(\mathbb{R})$ spanned by the set

$$S = \{1 + x + 2x^2 + x^3, 2 + 2x + 4x^2 + 2x^3, 1 - 2x + x^2 - 2x^3\}$$

a)

Decide whether or not S is linearly independent. If **not**, write the **second** polynomial as a linear combination of the others. If it *is* linearly independent, write 0 for both coefficients.

$$\underbrace{2 + 2x + 4x^2 + 2x^3}_{\text{polynomial 2}} = \boxed{2} \checkmark \underbrace{(1 + x + 2x^2 + x^3)}_{\text{polynomial 1}} + \boxed{0} \checkmark$$

$$\underbrace{(1 - 2x + x^2 - 2x^3)}_{\text{polynomial 3}}$$

Submit part

Your coefficients are correct. You were awarded **2** marks. 

You scored **2** marks for this part.

Score: 2/2 

Answered

b)

(i) Find a subset of S that is a basis for W . (Select the polynomials to include in the subset)

☒ $1 + x + 2x^2 + x^3$

☐ $2 + 2x + 4x^2 + 2x^3$

☒ $1 - 2x + x^2 - 2x^3$

(ii) Hence find the dimension of W :

Submit part

Correct. Dimension is 2 and polynomial 3 along with either 1 or 2 can make a basis. You were awarded 3 marks.



You scored 3 marks for this part.

Score: 3/3

Answered

c)

Find a basis for the **row space** of A . Enter your basis as a matrix comprised of your basis row vectors. You can increase the number of rows as required.

Rows: Columns:

1	2	0	1	0
0	0	1	-1	0
0	0	0	0	1

Submit part

Basis given for the row space of A is correct. You were awarded **1** mark. ✓

You scored **1** mark for this part.

Score: 1/1 ✓

Answered

Submit all parts

Score: 6/6 ✓

Try another question like this one

Reveal answers

The following two matrices are related by a sequence of elementary row operations:

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 3 & -1 & 1 & 1 & 3 \\ 2 & 1 & -2 & -1 & 7 \\ 2 & -2 & -3 & 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \end{pmatrix}$$

Select a subset of matrices from the following which can be used to give a basis that spans $\mathbf{M}_{2,2}(\mathbb{R})$?

☒ $\begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix}$ ☒ $\begin{pmatrix} 1 & -1 \\ 1 & -2 \end{pmatrix}$ ☒ $\begin{pmatrix} 1 & 1 \\ -2 & -3 \end{pmatrix}$ ☒ $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ ☐ $\begin{pmatrix} 1 & 3 \\ 7 & 0 \end{pmatrix}$ ☐ Not possible

Correct. The matrices selected will form a basis that spans the set of real 2x2 matrices. You were awarded 4 marks. 

You scored 4 marks for this part.

Score: 4/4 

Answered